

History of Computer: Charles Babbage, Early History, Different Generations, and Brief History

General



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History of Computer Overview

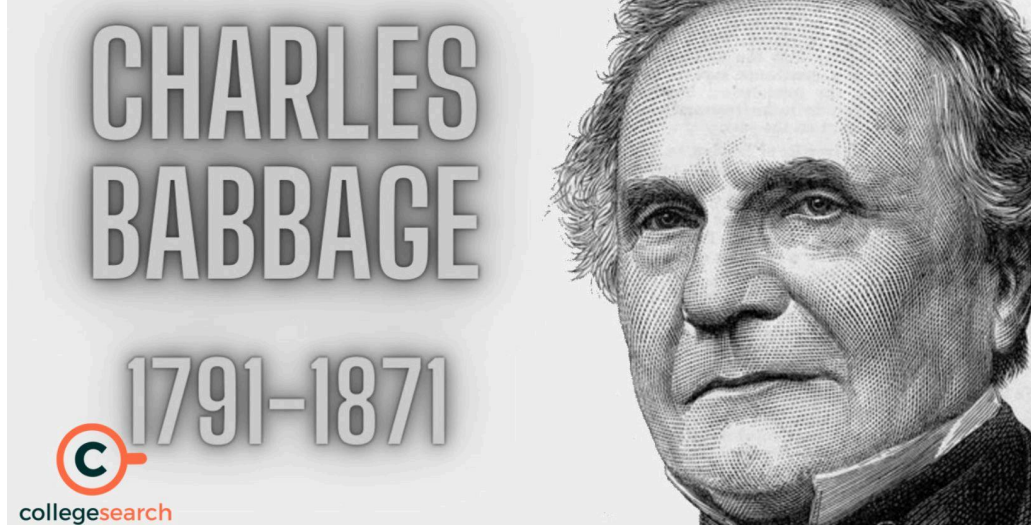
Prehistoric humans invented the earliest counting instrument. They counted using sticks, stones, and bones. More computer devices were produced over time as human intellect and technology advanced. While we now use computers for both work and entertainment, the computer was originally designed for a very different purpose. The computer has been around for a long time. Computers are classified into five generations. This leads to smaller, more powerful, and more resilient systems that are less expensive.

What is a Computer?

It is essential to understand the history of computers while studying the many elements of computing. Charles Babbage created the Analytical Engine, which was a type of universal computer. It aids us in comprehending the evolution and advancement of technology across time. It is also a topic covered in competitive and banking exams.

Who discovered Computers?

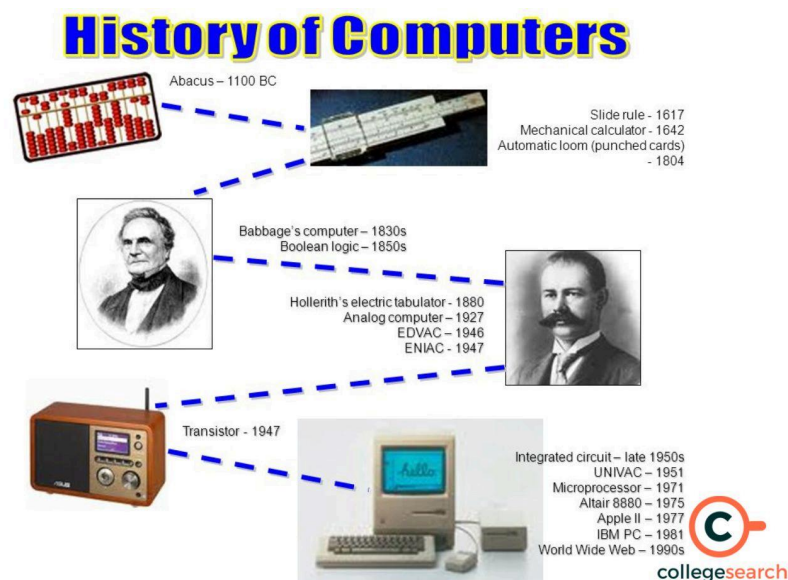
Charles Babbage, an English mathematician, and inventor, is credited with developing the first automated digital computer. Babbage created ideas for the Analytical Engine in the mid-1830s.



Read more about [Who Invented Computer?](#)

History of Computer

The term '**computer**' has an intriguing origin. It was initially used in the **16th century** to describe someone **who computed or did computations**. Until the twentieth century, the term was used in the same way as a noun. Women were employed to work as human computers, performing all types of calculations. By the late nineteenth century, the term was also used to denote devices that performed computations. The term is now often used to denote **programmable digital devices** that run on electricity.



Early History of Computer

For thousands of years, machines have been employed for computation since the development of humanity. The founder of computers, **Charles Babbage**, then began creating the **first mechanical computer in 1822**. Then, in **1833**, he created an **Analytical Engine**, which was a general purpose computer. It included an **ALU** (Arithmetic Logic Unit) and a **control unit**.

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A vintage computer system is shown, consisting of a CRT monitor, a base unit, and a keyboard. The monitor displays a circuit diagram titled "PROJECT" with various components and labels. The base unit is a horizontal, light-colored device with a floppy disk drive and a power switch. The keyboard is a standard QWERTY layout with a numeric keypad. A mouse is also visible on the right side of the keyboard. The background is a plain, light-colored wall.



Read more about the [First Computer in the World](#).

History of Computer: Different Generations

The developments of contemporary computers are sometimes referred to as the generation of computers in computer history. We are **now using computers from the fifth generation**. The following table gives the key aspects of these five computer generations-

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1st Generation	The initial generation of computers (1946-1959) was sluggish, large, and costly. Vacuum tubes were employed as the primary components of the CPU and memory in these computers. These computers relied heavily on batch operating systems and punch cards. In this generation, magnetic tape and paper tape were employed as output and input technologies, respectively.	EDVAC (Electronic Discrete Variable Automatic Computer) ENIAC (Electronic Numerical Integrator and Computer) UNIVACI (Universal Automatic Computer) IBM-650 IBM-701
2nd Generation	The transistor computer period spanned the second generation (1959-1965). These computers employed transistors, which were inexpensive, small, and required less power, making transistor computers quicker than first-generation computers. Magnetic cores were employed as primary memory in this generation, whereas magnetic discs and tapes were used as secondary storage. These computers employed assembly language and programming languages such as COBOL and FORTRAN, as well as batch processing and multiprogramming operating systems.	IBM 1620 IBM 7094 CDC 1604 CDC 3600 UNIVAC 1108
3rd Generation	Instead of transistors, third-generation computers employed integrated circuits (ICs). A single integrated circuit (IC) may contain a large number of	IBM-360 series Honeywell-6000 series PDP (Personal Data)
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	these machines employed remote processing, time-sharing, and multi-programming. This generation also employed high-level programming languages such as FORTRAN-II TO IV, COBOL, PASCAL PL/1, and ALGOL-68.	TDC-316
4th Generation	The fourth generation of computers (1971-1980) featured very large-scale integrated (VLSI) circuits, which consisted of a chip containing millions of transistors and other circuit parts. These processors made computers of this generation more small, powerful, quick, and economical. These computers utilized real-time, time-sharing, and a distributed operating system.	DEC 10 STAR 1000 PDP 11 CRAY-1(Super Computer) CRAY-X-MP(Super Computer)
5th Generation	VLSI technology was superseded by ULSI (Ultra Large Scale Integration) in fifth-generation (1980-to-date) computers. It enabled the manufacture of microprocessor chips with 10 million electrical components. Parallel processing hardware and AI (Artificial Intelligence) software were employed in this generation of computers. C, C++, Java,.Net, and other programming languages were employed in this generation.	Desktop Laptop NoteBook UltraBook Chromebook



Generation of Computer (1st to 5th)



Read more about [When was the Computer Invented?](#)

Brief History of Computer

The computer's creators had to recognize that their construction was more than just a number cruncher or a calculator. They had to deal with all of the challenges that come with creating such a machine, executing the concept, and actually manufacturing it. The history of computers is the tale of these problems being solved.

History of Computer: 19th Century

1801: In 1801, a **weaver and merchant from France named Joseph Marie Jacquard** invented a loom that used perforated wooden cards to mechanically weave fabric motifs.

1822: In 1822, **mathematician Charles Babbage** built the **first steam-powered calculating engine** capable of computing number tables. The "Difference Engine" concept failed due to a lack of available technology at the time.

1848. **Lovelace** also provides a **step-by-step explanation of using Babbage's machine** to compute Bernoulli numbers.

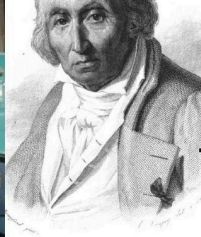
1890: **Inventor Herman Hollerith** developed the punch card system used to calculate the **1880 U.S. census**. He went on to start the company that would become IBM.

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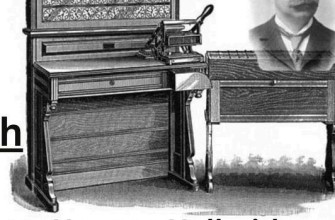


Joseph Marie Jacquard
invented a loom

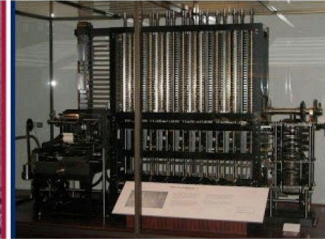


Lovelace provides the
step-by-step explanation
of using Babbage's
machine.

History of Computer: 19th Century.



Inventor Herman Hollerith
developed the punch card system



History of Computer: 20th Century

1930: **Vannevar Bush** devised and built the first large-scale automated general-purpose mechanical analog computer, the **Differential Analyzer**.

1936: **Alan Turing** devised the **Turing machine**, a **universal machine** that could calculate anything that could be computed.

1939: **Bill Hewlett** and **David Packard** discovered **Hewlett-Packard** in a garage in **Palo Alto, California**, in **1939**.

1941: **Konrad Zuse**, a **German inventor, and engineer**, finished his **Z3 machine**, the world's first digital computer, in 1941. The machine, however, was destroyed during a World War II bombing raid on **Berlin**. **J.V. Atanasoff** and graduate student **Clifford Berry** created a computer that can solve 29 equations at once. The first time data may be stored in a computer's main memory.

1945: **John Mauchly** and **J. Presper Eckert** of the **University of Pennsylvania** invent the **Electronic Numerical Integrator and Calculator (ENIAC)** in **1945**. It was Turing-complete and capable of solving "a vast class of numerical problems" by reprogramming, giving it the moniker "**Grandfather of Computers**."

1946: The **UNIVAC I (Universal Automatic Computer)** was the **first general-purpose electronic digital computer** created in the United States for business purposes in 1946.

1949: A team at the **University of Cambridge** develops the **Electronic Delay Storage Automatic Calculator (EDSAC)**, which is the "**first practical stored-program computer**."

1950: In Washington, DC, the **Standards Eastern Automatic Computer (SEAC)** was constructed, making it the first stored-program computer in the United States.



Differential Analyzer.

History of Computer: 20th Century



Alan Turing devised the Turing Machine.



Bill Hewlett and David Packard discovered Hewlett-Packard
Konrad Zuse, a German inventor

Electronic Numerical Integrator and Calculator (ENIAC) in 1945



History of Computer: Late 20th Century

1953: **Grace Hopper**, a computer scientist, develops the **first computer language, COBOL**, which stands for **Common Business-Oriented Language**. It allowed a computer user to provide instructions to the computer in English-like words rather than numbers.

1954: The **FORTRAN programming language**, an abbreviation for **FORmula TRANslation**, was established in 1954 by **John Backus** and a team of IBM programmers. Furthermore, IBM created the 650.

1958: **Jack Kilby** and **Robert Noyce** invented the **integrated circuit**, often known as the **computer chip**, in 1958.

1962: Atlas, the computer, first appears in 1962. It was the **world's fastest computer at the time**, and it pioneered the notion of "**virtual memory**."

1964: **Douglas Engelbart** offers a **contemporary computer prototype** with a mouse and a **graphical user interface (GUI)** in 1964.

1969: **Bell Labs developers** headed by **Ken Thompson** and **Dennis Ritchie** unveiled **UNIX**, an operating system written in C that solved program compatibility issues.

1970: Intel unveils the Intel 1103, the **first Dynamic Access Memory (DRAM) chip**.

1971: **Alan Shugart** and a team of IBM employees devised the **floppy disc in 1971**. In the same year, **Xerox created the first laser printer**, which not only made billions of dollars but also signaled the start of a new era in computer printing.

1973: **Robert Metcalfe**, a member of **Xerox's research department**, invented **Ethernet in 1973**, which is used to link multiple computers and other devices.

1976: **Steve Jobs** and **Steve Wozniak** establish **Apple Computers**, introducing the world to the **Apple I**, the **first computer with a single-circuit board**.

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1979: **WordStar**, a word processing program from MicroPro International, is published in 1979.

1983: The CD-ROM, which could hold 550 megabytes of pre-recorded data, was released in 1983. This year also saw the introduction of the Gavilan SC, the first portable computer with a flip-form design and the **first to be sold as a "laptop."**

1984: Apple introduced the **Macintosh during a Super Bowl XVIII ad in 1984**. It was \$2,500 in price.

1985: **Microsoft releases Windows**, which allows for **multitasking using a graphical user interface**. **C++**, a programming language, has also been released. Tim Berners-Lee, an English programmer, and scientist, developed HyperText Markup Language, often known as HTML, in 1990. In addition, he developed the term **"World Wide Web."** The first browser, a server, **HTML**, and **URLs** are all part of it.

1996: **Sergey Brin and Larry Page** founded the **Google search engine** at Stanford University in 1996.

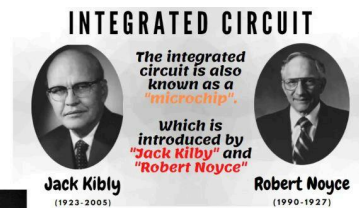
1988: **Apple debuts the iMac**, an all-in-one **Macintosh desktop computer**, in 1998. These **\$1,300 PCs** had a **4GB hard disk**, **32 MB of RAM**, a **CD-ROM**, and a **15-inch display**.

1999: **Wi-Fi**, an acronym for **"wireless fidelity,"** is invented in 1999, with an initial range of up to 300 feet.

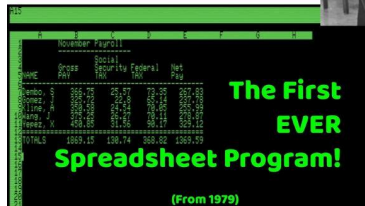


Grace Hopper develops the first computer language, COBOL,

History of Computer: 20th Century



Bell Labs developers, Ken Thompson and Dennis Ritchie



Douglas Engelbart

Computer mouse inventor

January 30, 1925 - July 2, 2013



History of Computer: Late 21st Century

2000: In the year 2000, the **USB flash drive** was first released. When used for data storage, they were faster and had greater storage space than other storage medium alternatives.

2001: **Apple introduces Mac OS X**, subsequently renamed OS X and then just macOS, as the replacement for the traditional Mac Operating System.

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Intel-based mobile computer from the firm.

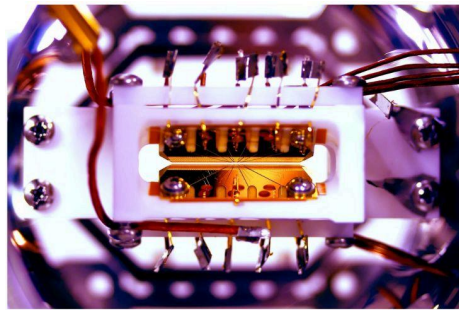
2009: Microsoft introduced **Windows 7** in 2009.

2011: Google releases the **Chromebook**, which runs **Google Chrome OS**, in 2011.

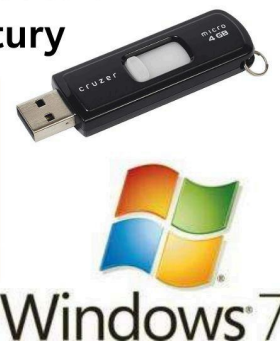
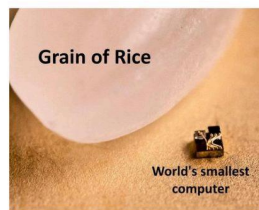
2014: The world's tiniest computer, the **University of Michigan Micro Mote (M3)**, was built in 2014.

2015: **Apple Watch** is introduced in 2015. Microsoft also introduced Windows 10.

2016: The world's first reprogrammable quantum computer is constructed in 2016.



History of Computer: Late 21st Century



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